

DATA STRUCTURES: DICTIONARIES

LEARNING OUTCOMES

At the end of this session, the students should be able to:

1. Use dictionaries in python; and
2. Create a simple program that uses a dictionary as the data structure.

CONTENT

- I. Dictionary
- II. Syntax
- III. Operations

DICTIONARY

- dictionaries (objects of type *dict*) are like lists except that “indices” may not be integers
- Key-value pairs
- Elements in a dictionary are not ordered
- Mutable

Syntax

In Python, a dictionary can be declared using the following syntax:

```
01 <dictionary_name> = {key:value, key:value...}
```

where:

key - any immutable object

value - any python object

Examples:

```
01 empty_dict = {}
02
03 numbers = { "one":1, "two":2, "three":3, "four":4, "five":5}
04
05 capitals = { "Philippines":"Manila",
06             "Japan":"Tokyo",
07             "South Korea":"Seoul",
08             "Australia":"Canberra",
09             "Canada":"Ottawa" }
10 numbers2 = { 1:"one", "two":2, "three":3, 4:"four", 5:"five"}
```

To access an element in a dictionary, we use the following syntax:

```
01 <dictionary_name>[key]
```

Examples:

```
01 numbers["one"]
02 numbers2[4]
03 print("The capital of Japan is", capital["Japan"])
```

```
04 print(capital[1]) #will produce an error
```

OPERATIONS

Add

To add an element in a dictionary, we use the following syntax:

```
01 <dictionary_name>[<new_key>] = <new_value>
```

Edit

Since all keys in a dictionary are unique, we can edit an item by “adding” using an old key

```
01 <dictionary_name>[<old_key>] = <new_value>
```

Delete

To delete an element in a dictionary, we use the following syntax:

```
01 del <dictionary_name>[<key>]
02 # or
03 <variable> = <dictionary_name>.pop(<key>)
```

To delete all entries in a dictionary, we use the following syntax:

```
01 <dictionary_name>.clear()
```

Check membership

To check if a certain key is already in a dictionary, we use the following syntax:

```
01 <key> in <dictionary_name> #will return True or False
```

Example:

```
01 if "Germany" in capitals:
02     print("Germany is already in the dictionary")
03 else:
04     print("Germany is not yet in the dictionary")
```

Get iterable objects

```
01 <dictionary_name>.keys() #get all the keys
02 <dictionary_name>.values() #get all the values
03 <dictionary_name>.items() #get all the items
```

Example:

```
01 for k in capitals:
02     print(k)
03
04 for k,v in capitals.items():
05     print(v, "is the capital of", k)
```

REFERENCES

JMJ Bawagan Teacher's Guide